

SIN'GYE, Korea, Democratic People's Republic ofWMO: **470670**Lat: **38.500N**Lon: **126.533E**Elev: **97**StdP: **100.17**Time Zone: **9.00 (E09)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.5	-13.2	-22.7	0.5	-12.8	-20.5	0.6	-10.7	8.5	-4.8	7.5	-6.0	1.0	320	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	31.6	24.5	30.1	23.9	28.9	23.1	26.3	29.9	25.6	28.6	24.8	27.3	1.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	20.8	28.3	24.7	19.9	27.3	24.1	19.2	26.4	82.8	30.1	79.3	28.8	76.2	27.4	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	6.6	5.5	-18.0	33.8	3.2	1.6	-20.4	34.9	-22.3	35.9	-24.1	36.8	-26.5	38.0		
(5)			-18.3	27.9	3.0	1.3	-20.5	28.9	-22.2	29.7	-23.9	30.4	-26.1	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-5.2	-1.9	4.0	10.5	16.9	21.6	24.3	24.7	19.9	13.1	5.2	-2.7
	DBStd	10.94	4.02	4.00	3.51	3.55	2.92	2.27	2.21	2.33	2.58	3.47	4.69	4.40
	HDD10.0	1591	473	334	188	36	0	0	0	0	0	13	154	394
	HDD18.3	3268	731	568	443	236	64	3	0	0	14	163	394	652
	CDD10.0	1929	0	0	3	51	213	348	443	455	297	109	11	0
	CDD18.3	563	0	0	0	1	18	101	184	197	61	2	0	0
	CDH23.3	3846	0	0	0	9	192	634	1190	1502	303	16	0	0
	CDH26.7	967	0	0	0	1	34	153	312	436	31	1	0	0
Wind	WSAvg	1.4	1.3	1.6	1.9	2.3	2.0	1.4	1.5	0.9	0.8	1.0	1.3	1.3
Precipitation	PrecAvg	1261	14	22	26	59	102	128	403	295	118	49	42	18
	PrecMax	2220	43	72	69	134	261	450	898	704	345	164	109	55
	PrecMin	509	1	1	2	4	20	36	127	38	7	4	4	2
	PrecStd	343	12	19	18	32	62	91	191	189	91	45	27	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	11.5	18.1	24.8	29.5	31.9	34.0	33.5	29.2	25.2	18.0	9.4
		MCWB	3.0	7.9	10.0	14.8	18.3	20.9	25.7	26.8	22.1	17.6	13.8	6.8
	2%	DB	3.7	8.1	14.9	21.8	27.4	30.1	31.2	31.8	27.6	22.9	15.6	7.2
		MCWB	1.0	4.6	8.8	12.9	17.3	20.7	25.0	25.7	21.1	16.4	12.0	5.3
	5%	DB	2.2	6.1	12.3	19.6	25.5	28.5	29.6	30.4	26.4	21.2	13.7	5.4
		MCWB	0.0	2.9	7.5	11.8	16.6	20.4	24.4	25.3	20.3	15.3	10.6	3.7
	10%	DB	0.9	4.2	10.2	17.4	23.5	27.1	28.3	29.1	25.2	19.5	12.1	3.6
		MCWB	-0.9	1.6	6.0	10.7	16.0	20.0	23.8	24.5	19.6	14.4	9.5	1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	9.1	11.8	16.9	20.6	24.1	27.5	28.0	24.1	19.1	15.3	7.8
		MCDB	5.1	10.0	14.7	21.1	26.5	28.1	31.3	31.6	26.4	22.9	17.0	8.7
	2%	WB	1.7	5.2	9.7	14.2	19.1	22.8	26.2	26.7	23.0	17.6	12.7	5.9
		MCDB	3.1	7.6	13.4	19.2	24.4	26.9	29.7	30.3	25.3	21.2	14.7	6.8
	5%	WB	0.4	3.3	8.0	12.8	18.0	21.9	25.3	26.0	22.0	16.4	11.2	3.8
		MCDB	1.8	5.4	11.9	17.9	22.9	25.8	28.3	29.3	25.0	19.8	13.4	5.2
	10%	WB	-0.8	1.8	6.6	11.6	17.0	21.2	24.7	25.3	21.0	15.1	9.8	2.0
		MCDB	0.8	4.0	9.7	16.0	21.9	24.9	27.1	28.0	24.0	18.7	11.6	3.3
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.5	9.8	11.1	11.0	9.1	6.4	7.3	9.7	11.3	8.8	8.3
		MCDBR	10.0	11.1	12.9	14.7	14.5	12.4	9.2	9.1	11.1	13.2	10.7	9.3
		MCWBR	7.9	7.9	7.8	7.3	6.4	5.2	4.3	4.3	5.4	7.1	7.3	7.4
	5% WB	MCDBR	8.6	9.5	11.2	12.2	11.6	9.4	7.0	7.6	8.3	11.1	9.3	8.5
		MCWBR	7.3	7.4	7.4	7.2	6.2	5.0	4.0	4.4	5.3	6.9	7.1	7.3
Clear-Sky Solar Irradiance	taub	0.415	0.484	0.556	0.619	0.627	0.705	0.676	0.575	0.504	0.511	0.483	0.421	
	taud	1.928	1.814	1.707	1.627	1.651	1.563	1.676	1.900	2.005	1.872	1.877	1.952	
	Ebn at Noon	706	706	700	688	695	641	656	716	742	673	627	663	
	Edh at Noon	138	178	218	251	249	272	242	189	160	165	146	127	
All-Sky Solar Radiation	RadAvg	2.50	3.20	4.19	4.88	5.49	5.14	3.94	4.28	4.20	3.52	2.37	2.16	
	RadStd	0.14	0.23	0.37	0.37	0.56	0.52	0.56	0.56	0.42	0.22	0.24	0.16	